

Final Exam Review

Example

- Let S denote the outwardly oriented piecewise-smooth surface consisting of
 - $S_1 \stackrel{\text{df}}{=} \{(x, y, z) \in \mathbb{R}^3 \mid \text{Closed double cone } z^2 = x^2 + y^2 \text{ and } \text{Above the } xy\text{-plane } 0 \leq z \leq 1\}$ and
 - $S_2 \stackrel{\text{df}}{=} \{(x, y, z) \in \mathbb{R}^3 \mid \text{Cylinder } x^2 + y^2 = 1 \text{ and } 1 \leq z \leq 4\}$.
- Define a smooth space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (y, -x, z).$$

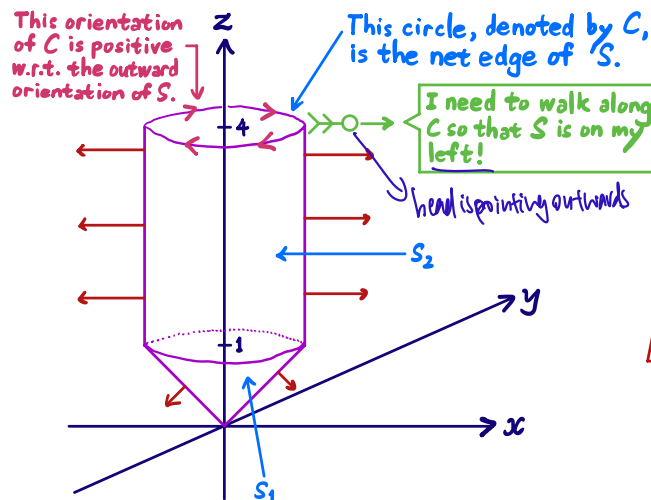
Stokes' Theorem

$$\iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

↓
edge of S

Problem: Use Stokes' Theorem to find the flux of $\text{curl}(\mathbf{F})$ across S .

Solution



- Define a vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ by $\vec{r}(t) \stackrel{\text{df}}{=} \langle \cos(t), \sin(t), 4 \rangle$.
- Let C be equipped with the orientation induced by $\vec{r}$, which is counterclockwise when viewed from above. **Bummer!**
- This orientation of C is opposite to the positive orientation of C w.r.t. the outward orientation of S , which is clockwise when viewed from above, as depicted in the figure to the left.
- Therefore, by Stokes' Theorem,

$$\begin{aligned} \iint_S \text{curl}(\mathbf{F}) \cdot d\mathbf{S} &= \ominus \int_C \mathbf{F} \cdot d\mathbf{r} \\ &= - \int_0^{2\pi} \mathbf{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \end{aligned}$$

Final Exam Review

$$\begin{aligned} &= - \int_0^{2\pi} \vec{F}(\cos(t), \sin(t), 4) \cdot \langle -\sin(t), \cos(t), 0 \rangle dt \\ &= - \int_0^{2\pi} \langle \sin(t), -\cos(t), 4 \rangle \cdot \langle -\sin(t), \cos(t), 0 \rangle dt \\ &= - \int_0^{2\pi} [-\sin^2(t) - \cos^2(t)] dt \\ &= \int_0^{2\pi} [\sin^2(t) + \cos^2(t)] dt \\ &= \int_0^{2\pi} 1 dt \\ &= [t]_0^{2\pi} \\ &= 2\pi. \otimes \end{aligned}$$

Alternative solution

- Define a vector function $\vec{r}: [0, 2\pi] \rightarrow \mathbb{R}^3$ by

$$\vec{r}(t) \stackrel{\text{df}}{=} \langle \cos(-t), \sin(-t), 4 \rangle = \langle \cos(t), -\sin(t), 4 \rangle.$$

- Let C be equipped with the orientation induced by $\vec{r}$, which, this time, is clockwise when viewed from above.
- This orientation of C is positive ^{Fantastic!} w.r.t. the outward orientation of S .
- Therefore, by Stokes' Theorem,

$$\begin{aligned} \iint_S \text{curl}(\vec{F}) \cdot d\vec{S} &= \oint_C \vec{F} \cdot d\vec{r} \\ &= \int_0^{2\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\ &= \int_0^{2\pi} \langle -\sin(t), -\cos(t), 4 \rangle \cdot \langle -\sin(t), -\cos(t), 0 \rangle dt \\ &= \int_0^{2\pi} [\sin^2(t) + \cos^2(t)] dt \\ &= \int_0^{2\pi} 1 dt \end{aligned}$$

Final Exam Review

$$\begin{aligned} &= [t]_0^{2\pi} \\ &= 2\pi. \times \end{aligned}$$

Example

- Let C denote the triangle in $\mathbb{R}^3$ with vertices

$$(3, 0, 0), \quad (0, 3, 0), \quad \text{and} \quad (0, 0, 3),$$

oriented so that its projection onto the xy -plane is a positively oriented triangle in $\mathbb{R}^2$.

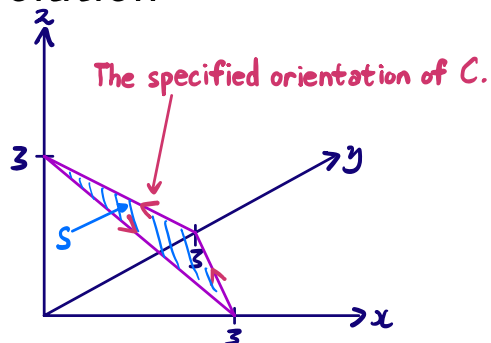
- Define a smooth space vector field $\mathbf{F}$ on $\mathbb{R}^3$ by

$$\mathbf{F}(x, y, z) \stackrel{\text{df}}{=} (x + y^2, y + z^2, z + x^2).$$

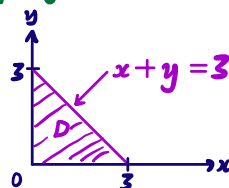
Problem: Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.

$$\iint_S \text{curl}(\vec{F}) \cdot d\vec{S}$$

Solution



- The triangle C lies entirely on the plane Π given by the equation $x + y + z = 3$.
- Let S denote the closed triangular region of Π that is enclosed by C .
- Let D denote the projection of S onto the xy -plane:



Final Exam Review

- Then $D = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 3 \text{ and } 0 \leq y \leq 3 - x\}$.
- Furthermore,
$$S = \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in D \text{ and } x + y + z = 3\}$$
$$= \{(x, y, 3 - x - y) \in \mathbb{R}^3 \mid (x, y) \in D\}.$$
- Define a vector function $\vec{r}: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $\vec{r}(x, y) \stackrel{\text{df}}{=} \langle x, y, 3 - x - y \rangle$.
- For all $x, y \in \mathbb{R}$, we have

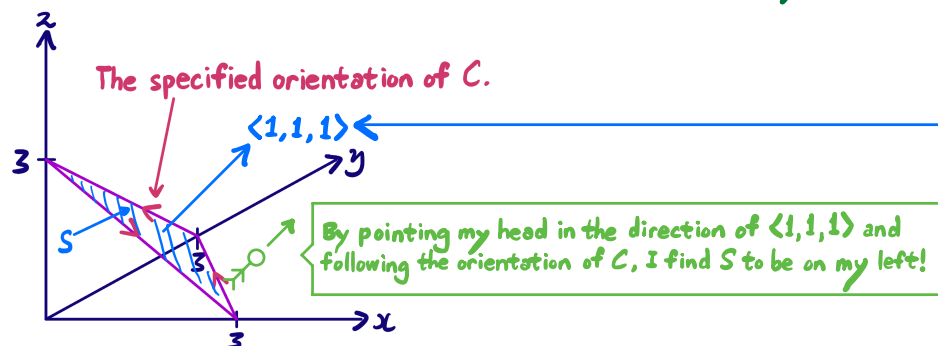
$$\frac{\partial \vec{r}}{\partial x}(x, y) = \langle 1, 0, -1 \rangle,$$

$$\frac{\partial \vec{r}}{\partial y}(x, y) = \langle 0, 1, -1 \rangle,$$

$$\left(\frac{\partial \vec{r}}{\partial x} \times \frac{\partial \vec{r}}{\partial y} \right)(x, y) = \langle 1, 1, 1 \rangle.$$

Points away from the origin.

- Then $(\vec{r}, D)$ is a nice surface parametrization of S .
- Equip S with the orientation induced by $(\vec{r}, D)$.



Fantastic!

- By the Right-Hand Grip Rule, the orientation of C is positive w.r.t. this orientation of S .

Final Exam Review

- Now, for all $x, y, z \in \mathbb{R}$, we have

$$\begin{aligned} [\text{curl}(\vec{F})](x, y, z) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y^2 & y+z^2 & z+x^2 \end{vmatrix} \\ &= \langle -2z, -2x, -2y \rangle. \end{aligned}$$

- Therefore, by Stokes' Theorem,

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \iint_S \text{curl}(\vec{F}) \cdot d\vec{S} \\ &= \iint_D [\text{curl}(\vec{F})](\overset{x}{x}, \overset{y}{y}, \overset{z}{3-x-y}) \cdot \langle 1, 1, 1 \rangle dA(x, y) \\ &= \iint_D \langle -2(3-x-y), -2x, -2y \rangle \cdot \langle 1, 1, 1 \rangle dA(x, y) \\ &= \iint_D \langle -6+2x+2y, -2x, -2y \rangle \cdot \langle 1, 1, 1 \rangle dA(x, y) \\ &= \iint_D (-6) dA(x, y) \end{aligned}$$

$$\int_0^3 \left[\int_0^{3-x} (-6) dy \right] dx$$

= ...

$$\begin{aligned} &-6 \iint_D 1 dA(x, y) \\ &= -6 \cdot \text{Area}(D) \\ &= -6 \cdot \frac{1}{2}(3)(3) \\ &= -27. \otimes \otimes \end{aligned}$$

-6 can be moved outside the double integral ONLY BECAUSE it is a constant. In general, you CANNOT do this and must evaluate an iterated double integral.

Example

- Let C denote the negatively oriented triangle in $\mathbb{R}^2$ with vertices

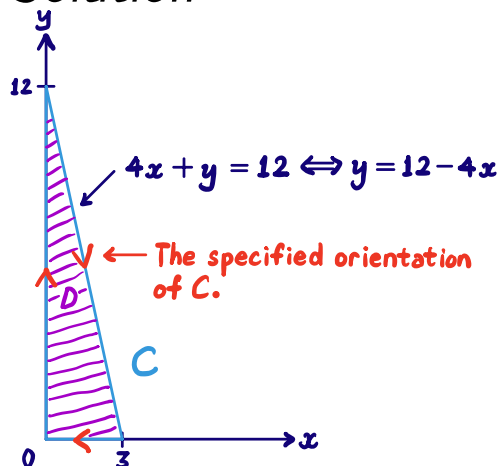
$$(0, 0), \quad (0, 12), \quad \text{and} \quad (3, 0).$$

- Define a smooth plane vector field on $\mathbb{R}^2$ by

$$\mathbf{F}(x, y) \stackrel{\text{df}}{=} (y \overset{P(x,y)}{\cos(x)} - xy \sin(x), xy \overset{Q(x,y)}{+} x \cos(x)).$$

Problem: Use Green's Theorem to evaluate $\int_C \mathbf{F} \bullet d\mathbf{r}$.

Solution



- Let D denote the closed region in $\mathbb{R}^2$ that is enclosed by C .
- Note that D is a Type-I region in $\mathbb{R}^2$ with the specification $\{(x, y) \in \mathbb{R}^2 \mid 0 \leq x \leq 3 \text{ and } 0 \leq y \leq 12 - 4x\}$.
- Let $\vec{F} = \langle P, Q \rangle$.
- Then, for all $x, y \in \mathbb{R}$, we have

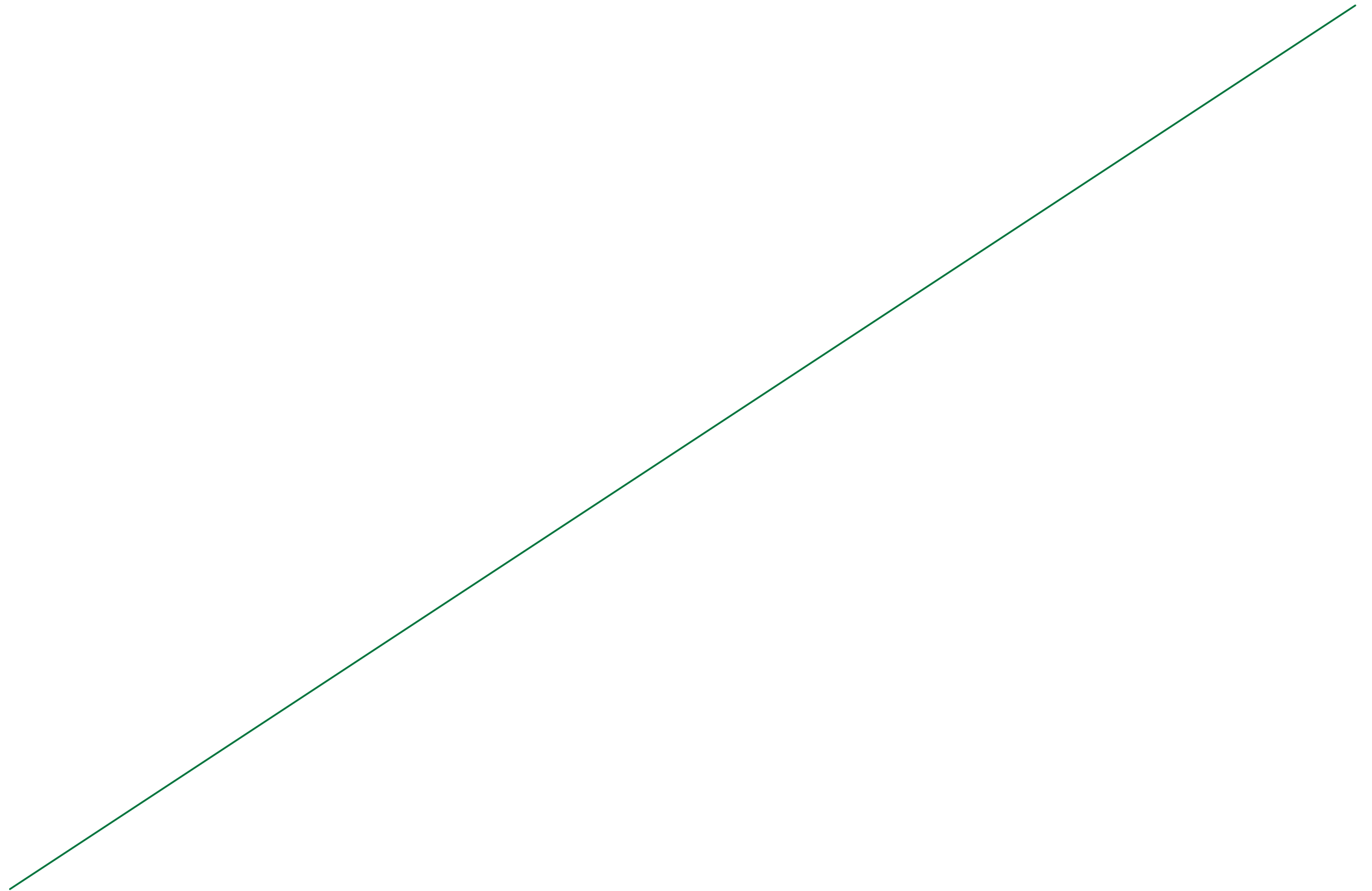
$$\begin{aligned} \frac{\partial Q}{\partial x}(x, y) &= y + \cos(x) - x \sin(x), \\ \frac{\partial P}{\partial y}(x, y) &= \cos(x) - x \sin(x). \end{aligned}$$

Final Exam Review

- Therefore, by Green's Theorem, Apply Green's Theorem to this.

$$\begin{aligned} \int_{\odot} \vec{F} \cdot d\vec{r} &= - \int_{\ominus} \vec{F} \cdot d\vec{r} \\ \text{Negatively oriented} & \quad \text{Positively oriented} \\ &= - \iint_D \left[\frac{\partial Q}{\partial x}(x,y) - \frac{\partial P}{\partial y}(x,y) \right] dA(x,y) \\ &= - \iint_D y \, dA(x,y) \\ &= - \int_0^3 \left[\int_0^{12-4x} y \, dy \right] dx \\ &= - \int_0^3 \left[\frac{1}{2} y^2 \right]_{y=0}^{y=12-4x} dx \\ &= - \int_0^3 \frac{1}{2} (12-4x)^2 dx \\ &= - \int_0^3 8(3-x)^2 dx \\ &= - 8 \int_0^3 (3-x)^2 dx \\ &= - 8 \left[-\frac{1}{3} (3-x)^3 \right]_0^3 \\ &= - 8 \cdot \frac{1}{3} (3)^3 \\ &= - 8 \cdot 9 \\ &= - 72. \otimes \end{aligned}$$

Final Exam Review



Example

Let C denote the oriented curve in $\mathbb{R}^2$ having the orientation-reversing parametrization $\mathbf{r} : [0, 1] \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{r}(t) \stackrel{\text{df}}{=} \left(e^{t^2-t} - \cos(2\pi t), 2 \sin\left(\frac{\pi t^2}{2}\right) - t^9 \right).$$

Problem: Evaluate $\int_C \left[\overbrace{(2xy + e^{-x^2})}^{P(x,y)} dx + \overbrace{\left(x^2 + y \cos\left(\frac{\pi y^2}{2}\right)\right)}^{Q(x,y)} dy \right]$.

Solution

- As $\vec{r}$ is an orientation-reversing parametrization of C , the start point of C is $\vec{r}(1) = (0, 1)$ (instead of $\vec{r}(0)$) and the end point of C is $\vec{r}(0) = (0, 0)$ (instead of $\vec{r}(1)$).

- Define a plane vector field $\vec{F} = \langle P, Q \rangle$ on $\mathbb{R}^2$ by

$$\vec{F}(x, y) \stackrel{\text{df}}{=} \left\langle 2xy + e^{-x^2}, x^2 + y \cos\left(\frac{\pi y^2}{2}\right) \right\rangle.$$

- For all $x, y \in \mathbb{R}$, we have

$$\frac{\partial Q}{\partial x}(x, y) = 2x = \frac{\partial P}{\partial y}(x, y).$$

- Hence, as $\mathbb{R}^2$ is an open and simply connected subset of $\mathbb{R}^2$, we find that $\vec{F}$ is conservative.

- We will now evaluate $\int_C \vec{F} \cdot d\vec{r}$ in two different ways.

The line integral that the problem is asking us to evaluate.

Final Exam Review

Solution 1

- As $\vec{F}$ is conservative, it has — by definition — a potential function f .
- For all $x, y \in \mathbb{R}$, we have $(\nabla f)(x, y) = \vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$, so

$$f_x(x, y) = 2xy + e^{-x^2} \quad \text{and} \quad f_y(x, y) = x^2 + y \cos\left(\frac{\pi y^2}{2}\right).$$

- Let F be any anti-derivative of the function from $\mathbb{R}$ to $\mathbb{R}$ that maps x to e^{-x^2} .
- It is NOT possible to express F in terms of the elementary functions (you will have been told this in Calculus II).
- This will not be a problem for us, however, as we shall see later.
- Now, there exist functions $c: \mathbb{R} \rightarrow \mathbb{R}$ and $d: \mathbb{R} \rightarrow \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$,

Perform $\int \square dx$.

$$f(x, y) = x^2 y + F(x) + c(y),$$

$$f(x, y) = x^2 y + \frac{1}{\pi} \sin\left(\frac{\pi y^2}{2}\right) + d(x).$$

Perform $\int \square dy$.

- We may thus choose f so that for all $x, y \in \mathbb{R}$,

$$f(x, y) = x^2 y + \frac{1}{\pi} \sin\left(\frac{\pi y^2}{2}\right) + F(x).$$

⤵ hand together, we just match them

- Therefore, by the Fundamental Theorem of Line Integrals,

$$\int_C \vec{F} \cdot d\vec{r} = f(\text{end point of } C) - f(\text{start point of } C)$$

$$= f(0, 0) - f(0, 1)$$

$$= F(0) - \left[\frac{1}{\pi} + F(0)\right]$$

$$= -\frac{1}{\pi} \cdot \otimes$$

⚠ Remember that F is orientation reversing

Final Exam Review

Solution 2

- As $\vec{F}$ is conservative, it has the Path-Independence Property.
- Let $\mathcal{L}$ denote the directed line segment from $\underbrace{(0,1)}_{\text{Start point of } \mathcal{L}}$ to $\underbrace{(0,0)}_{\text{End point of } \mathcal{L}}$.
- The standard affine parametrization of $\mathcal{L}$ is the vector function $\vec{s}: [0,1] \rightarrow \mathbb{R}^2$ defined by

$$\begin{aligned}\vec{s}(t) &\stackrel{\text{def}}{=} \langle 0,1 \rangle + t \cdot [\langle 0,0 \rangle - \langle 0,1 \rangle] \\ &= \langle 0,1 \rangle + t \cdot \langle 0,-1 \rangle \\ &= \langle 0,1-t \rangle.\end{aligned}$$

- Therefore,

$$\begin{aligned}\int_{\mathcal{L}} \vec{F} \cdot d\vec{r} &= \int_{\mathcal{L}} \vec{F} \cdot d\vec{r} \quad (\text{by the Path-Independence Property}) \\ &= \int_0^1 \vec{F}(\vec{s}(t)) \cdot \vec{s}'(t) \, dt \\ &= \int_0^1 \vec{F}(0,1-t) \cdot \langle 0,-1 \rangle \, dt \\ &= \int_0^1 \left\langle \cancel{0}, (1-t) \cos\left(\frac{\pi(1-t)^2}{2}\right) \right\rangle \cdot \langle 0,-1 \rangle \, dt \\ &= \int_0^1 \left[-(1-t) \cos\left(\frac{\pi(1-t)^2}{2}\right) \right] \, dt \\ &= \left[\frac{1}{\pi} \sin\left(\frac{\pi(1-t)^2}{2}\right) \right]_0^1 \\ &= 0 - \frac{1}{\pi} \\ &= -\frac{1}{\pi} \cdot \otimes\end{aligned}$$

Example

Let R denote the closed region in the right half-plane $\mathbb{R}_{\geq 0} \times \mathbb{R}$ that is bounded by the four curves having the following equations:

$$\boxed{\frac{y}{x} + x} = 1, \quad \frac{y}{x} + x = 2, \quad \boxed{\frac{y}{x} - x} = -2, \quad \text{and} \quad \frac{y}{x} - x = 0.$$

u v

Problem: Use a change of variables to evaluate $\iint_R \frac{1}{x} \sin\left(\frac{y}{x} + x\right) dA(x, y)$.

Solution

- Let $\mathcal{O}$ denote the region in the (u, v) -plane specified by $\{(u, v) \in \mathbb{R}^2 \mid u > v\}$.
- Define functions $U: \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$ and $V: \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$ by $U(x, y) \triangleq \frac{y}{x} + x$ and $V(x, y) \triangleq \frac{y}{x} - x$.
- Then (U, V) is a continuous bijection from $\mathbb{R}_{>0} \times \mathbb{R}$ to $\mathcal{O}$ whose inverse, which we shall denote by T , is also continuous.
- We do not need to show this, however.
- Note that $T^{-1}[R] = [1, 2] \times [-2, 0]$.

Final Exam Review

- For all $x \in \mathbb{R}_{>0}$ and $y \in \mathbb{R}$, we have

$$\begin{aligned} |J_{T^{-1}}(x,y)| &= \left| \det \begin{pmatrix} U_x(x,y) & U_y(x,y) \\ V_x(x,y) & V_y(x,y) \end{pmatrix} \right| \\ &= \left| \det \begin{pmatrix} 1 - \frac{y}{x^2} & \frac{1}{x} \\ -1 - \frac{y}{x^2} & \frac{1}{x} \end{pmatrix} \right| \\ &= \left| \frac{1}{x} \left(1 - \frac{y}{x^2} \right) - \frac{1}{x} \left(-1 - \frac{y}{x^2} \right) \right| \\ &= \left| \frac{1}{x} - \frac{y}{x^3} + \frac{1}{x} + \frac{y}{x^3} \right| \\ &= \left| \frac{2}{x} \right| \\ &= \frac{2}{x} > 0. \end{aligned}$$

- Therefore,

$$\begin{aligned} \iint_R \frac{1}{x} \sin\left(\frac{y}{x} + x\right) dA(x,y) &= \iint_R \frac{\left[\frac{1}{x} \sin\left(\frac{y}{x} + x\right)\right]}{\left(\frac{2}{x}\right)} \cdot \frac{2}{x} dA(x,y) \\ &= \iint_R \boxed{\frac{1}{2} \sin\left(\frac{y}{x} + x\right)} \cdot \frac{2}{x} dA(x,y) \\ &\quad \text{Express in terms of } U(x,y) \text{ and } V(x,y). \\ &= \iint_R \frac{1}{2} \sin(U(x,y)) \cdot \boxed{\frac{2}{x} dA(x,y)} \\ &= \iint_{T^{-1}[R]} \frac{1}{2} \sin(u) \boxed{dA(u,v)} \\ &= \iint_{[1,2] \times [-2,0]} \frac{1}{2} \sin(u) dA(u,v) \\ &= \frac{1}{2} \iint_{[1,2] \times [-2,0]} \sin(u) dA(u,v) \end{aligned}$$

Final Exam Review

$$\begin{aligned} &= \frac{1}{2} \cdot \left[\int_1^2 \sin(u) \, du \right] \cdot \left(\int_{-2}^0 1 \, dv \right) \\ &= \frac{1}{2} \cdot [-\cos(u)]_1^2 \cdot [v]_{-2}^0 \\ &= \frac{1}{2} \cdot [\cos(1) - \cos(2)] \cdot 2 \\ &= \cos(1) - \cos(2). \otimes \end{aligned}$$

Example

Let B be a closed rectangular box in $\mathbb{R}^3$ that satisfies the following two properties:

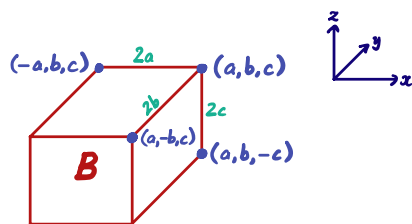
- Every edge of B is parallel to a coordinate axis.
- The eight vertices of B lie on the ellipsoid given by the equation

$$2x^2 + 72y^2 + 18z^2 = 288.$$

Problem: Use Lagrange multipliers to find the maximum possible volume of B .

Solution

- If (a,b,c) is a vertex of B , then the volume of B is $8|abc|$.



- Define functions $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ and $g: \mathbb{R}^3 \rightarrow \mathbb{R}$ by
$$f(x,y,z) \triangleq 8xyz,$$
$$g(x,y,z) \triangleq 2x^2 + 72y^2 + 18z^2.$$

- For all $x,y,z \in \mathbb{R}$, we have

$$(\nabla f)(x,y,z) = \langle 8yz, 8xz, 8xy \rangle,$$

$$(\nabla g)(x,y,z) = \langle 4x, 144y, 36z \rangle.$$

Final Exam Review

- Goal: To maximize f subject to the constraint $g(x,y,z) = 288$.
- This goal can be realized because of the following two facts:
 - (i) The ellipsoid is a bounded and closed subset of $\mathbb{R}^3$.
 - (ii) f is continuous.
- Now, for all $x, y, z, \lambda \in \mathbb{R}$, we have

$$\begin{cases} 2x^2 + 72y^2 + 18z^2 = 288 \\ 8yz = 4\lambda x \\ 8xz = 144\lambda y \\ 8xy = 36\lambda z \end{cases} \iff \begin{cases} \lambda \neq 0 \\ 2x^2 + 72y^2 + 18z^2 = 288 \\ 8yz = 4\lambda x \\ 8xz = 144\lambda y \\ 8xy = 36\lambda z \end{cases} \quad \text{or}$$

Note that $\lambda \neq 0$ forces $x, y, z \neq 0$.

$$\iff \begin{cases} \lambda \neq 0 \\ 2x^2 + 72y^2 + 18z^2 = 288 \\ 8xyz = 4\lambda x^2 \\ 8xyz = 144\lambda y^2 \\ 8xyz = 36\lambda z^2 \end{cases}$$

$$\iff \begin{cases} \lambda \neq 0 \\ 2x^2 + 72y^2 + 18z^2 = 288 \\ 8xyz = 4\lambda x^2 \\ 4\lambda x^2 = 144\lambda y^2 \\ 4\lambda x^2 = 36\lambda z^2 \end{cases}$$

$$\iff \begin{cases} x^2 + 36y^2 + 9z^2 = 144 \\ 2yz = \lambda x \\ x^2 = 36y^2 \\ x^2 = 9z^2 \end{cases}$$

$$\iff \begin{cases} 3x^2 = 144 \\ y^2 = \frac{x^2}{36} \\ z^2 = \frac{x^2}{9} \\ 2yz = \lambda x \end{cases}$$

$$\begin{cases} \lambda = 0 \\ 2x^2 + 72y^2 + 18z^2 = 288 \\ 8yz = 4\lambda x \\ 8xz = 144\lambda y \\ 8xy = 36\lambda z \end{cases}$$

$$\iff \begin{cases} \lambda = 0 \\ 2x^2 + 72y^2 + 18z^2 = 288 \\ 8yz = 0 \\ 8xz = 0 \\ 8xy = 0 \end{cases}$$

$$\iff \begin{cases} \lambda = 0 \\ x^2 + 36y^2 + 9z^2 = 144 \\ yz = 0 \\ xz = 0 \\ xy = 0 \end{cases}$$

$$\iff \begin{cases} \lambda = 0 \\ x = 0 \\ y = 0 \\ z^2 = 16 \end{cases} \quad \text{or} \quad \begin{cases} \lambda = 0 \\ x = 0 \\ z = 0 \\ y^2 = 4 \end{cases} \quad \text{or} \quad \begin{cases} \lambda = 0 \\ y = 0 \\ z = 0 \\ x^2 = 144 \end{cases}$$

$$\iff (x, y, z, \lambda) \in \{(0, 0, \pm 4, 0), (0, \pm 2, 0, 0), (\pm 12, 0, 0, 0)\}.$$

These solutions can be discarded as being irrelevant to the physical nature of the problem.

If this were a mathematical problem, then you would have to take these solutions into account.

Final Exam Review

$$\Leftrightarrow \begin{cases} x^2 = 48 \\ y^2 = \frac{4}{3} \\ z^2 = \frac{48}{9} \\ 2yz = \lambda x \end{cases}$$

$$\Leftrightarrow (x, y, z, \lambda) \in \left\{ \left(\pm 4\sqrt{3}, \pm \frac{2}{\sqrt{3}}, \pm \frac{4}{\sqrt{3}}, \pm \frac{4}{3\sqrt{3}} \right), \left(\pm 4\sqrt{3}, \pm \frac{2}{\sqrt{3}}, \mp \frac{4}{\sqrt{3}}, \mp \frac{4}{3\sqrt{3}} \right), \right. \\ \left. \left(\pm 4\sqrt{3}, \mp \frac{2}{\sqrt{3}}, \pm \frac{4}{\sqrt{3}}, \mp \frac{4}{3\sqrt{3}} \right), \left(\pm 4\sqrt{3}, \mp \frac{2}{\sqrt{3}}, \mp \frac{4}{\sqrt{3}}, \pm \frac{4}{3\sqrt{3}} \right) \right\}.$$

- The minimum value of f subject to the constraint $g(x, y, z) = 288$ is $-8(4\sqrt{3})\left(\frac{2}{\sqrt{3}}\right)\left(\frac{4}{\sqrt{3}}\right) = -\frac{256}{\sqrt{3}}$.
- The maximum value of f subject to the constraint $g(x, y, z) = 288$ is $8(4\sqrt{3})\left(\frac{2}{\sqrt{3}}\right)\left(\frac{4}{\sqrt{3}}\right) = \frac{256}{\sqrt{3}}$.
- Conclusion: The maximum possible volume of B is $\frac{256}{\sqrt{3}} \cdot \otimes$

Example

Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) \stackrel{\text{df}}{=} x^4 - 2x^2 + y^3 - 3y$.

Problem: Find all the local minima, local maxima, and saddle points of f .

Solution

• For all $x, y \in \mathbb{R}$, we have

$$\begin{aligned}f_x(x, y) &= 4x^3 - 4x = 4x(x+1)(x-1), \\f_y(x, y) &= 3y^2 - 3 = 3(y+1)(y-1),\end{aligned}$$

so

$$\begin{aligned}f_{xx}(x, y) &= 12x^2 - 4, \\f_{yy}(x, y) &= 6y, \\f_{xy}(x, y) &= 0, \\f_{yx}(x, y) &= 0.\end{aligned}$$

• For all $x, y \in \mathbb{R}$, we also have

$$\begin{aligned}(\nabla f)(x, y) = \langle 0, 0 \rangle &\iff \begin{cases} f_x(x, y) = 0 \\ f_y(x, y) = 0 \end{cases} \\&\iff \begin{cases} 4x(x+1)(x-1) = 0 \\ 3(y+1)(y-1) = 0 \end{cases}\end{aligned}$$

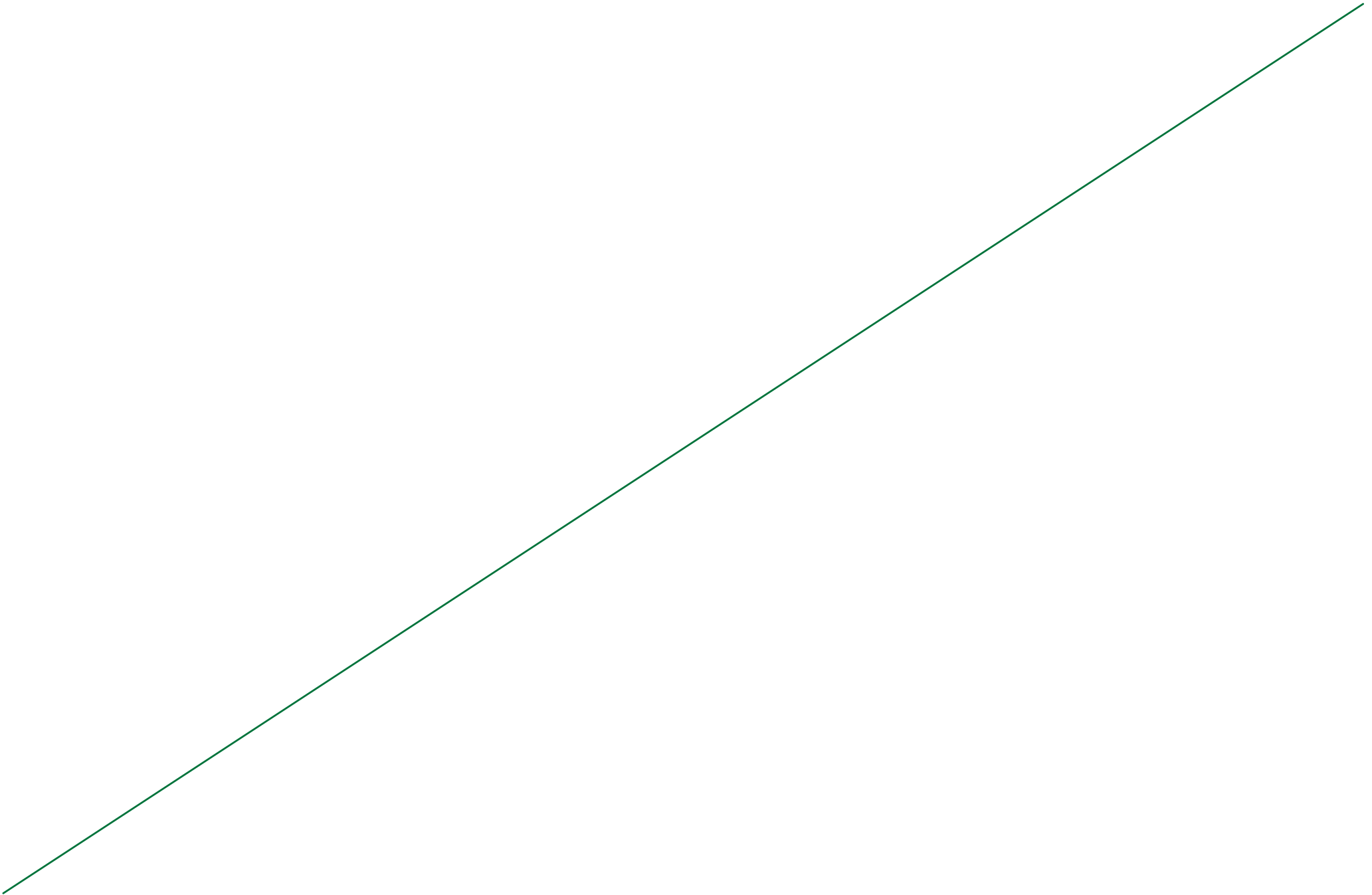
Final Exam Review

$$\Leftrightarrow \begin{cases} x = 0 \text{ or } x = \pm 1 \\ y = \pm 1 \end{cases}.$$

- | Critical point (x,y) | $f_{xx}(x,y) f_{yy}(x,y) - [f_{xy}(x,y)]^2$ | $f_{xx}(x,y)$ | Remark | Value $f(x,y)$ |
|------------------------|---|---------------|---------------|----------------|
| $(0,1)$ | — | NA | Saddle point | -2 |
| $(0,-1)$ | + | — | Local maximum | 2 |
| $(1,1)$ | + | + | Local minimum | -3 |
| $(1,-1)$ | — | NA | Saddle point | 1 |
| $(-1,1)$ | + | + | Local minimum | -3 |
| $(-1,-1)$ | — | NA | Saddle point | 1 |



Final Exam Review



Example

Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} \begin{cases} \frac{x^4 + y^4}{\sqrt{x^2 + y^2}}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

Problem: Determine whether or not f is totally differentiable at $(0, 0)$.

Solution

• For all $x \in \mathbb{R} \setminus \{0\}$, we have

$$f(x, 0) = \frac{x^4 + 0^4}{\sqrt{x^2 + 0^2}} = \frac{x^4}{\sqrt{x^2}} = \frac{x^4}{|x|} = \frac{|x|^4}{|x|} = |x|^3.$$

• As $f(0, 0) = 0$, we thus have, for all $x \in \mathbb{R}$, that $f(x, 0) = |x|^3$.

• Hence,

$$\begin{aligned} f_x(0, 0) &= \frac{d}{dx} [|x|^3] \Big|_{x=0} \\ &= \lim_{x \rightarrow 0} \frac{|x|^3 - 0}{x - 0} \\ &= \lim_{x \rightarrow 0} \frac{|x|^3}{x} \\ &= \lim_{x \rightarrow 0} |x|^2 \cdot \frac{|x|}{x} = 0. \end{aligned}$$

Bounded in absolute value by 1.

By the Squeeze Theorem.

Final Exam Review

- For all $y \in \mathbb{R} \setminus \{0\}$, we have

$$f(0, y) = \frac{0^4 + y^4}{\sqrt{0^2 + y^2}} = \frac{y^4}{\sqrt{y^2}} = \frac{y^4}{|y|} = \frac{|y|^4}{|y|} = |y|^3.$$

- As $f(0, 0) = 0$, we thus have, for all $y \in \mathbb{R}$, that $f(0, y) = |y|^3$.

- Similarly, $f_y(0, 0) = 0$.

- It follows that the tangent-plane function of f at $(0, 0)$, denoted by $L_{f, (0, 0)}$, exists.

- For all $x, y \in \mathbb{R}$, we have

$$\begin{aligned} L_{f, (0, 0)}(x, y) &= f(0, 0) + f_x(0, 0) \cdot (x - 0) + f_y(0, 0) \cdot (y - 0) \\ &= 0 + 0 \cdot x + 0 \cdot y \\ &= 0. \end{aligned}$$

- It now suffices to determine whether $\lim_{(x, y) \rightarrow (0, 0)} \frac{|f(x, y) - L_{f, (0, 0)}(x, y)|}{\|(x, y) - (0, 0)\|} = 0$.

- Let $(x, y) \in \mathbb{R}^2 \setminus \{(0, 0)\}$.

- Then

This is precisely what it means for f to be totally differentiable at $(0, 0)$.

$$\begin{aligned} \frac{|f(x, y) - L_{f, (0, 0)}(x, y)|}{\|(x, y) - (0, 0)\|} &= \frac{|f(x, y) - 0|}{\|(x, y)\|} \\ &= \frac{|f(x, y)|}{\|(x, y)\|} \\ &= \frac{\left(\frac{x^4 + y^4}{\sqrt{x^2 + y^2}} \right)}{\sqrt{x^2 + y^2}} \\ &= \frac{x^4 + y^4}{x^2 + y^2}. \end{aligned}$$

- Let (r, θ) be a polar-coordinate pair for (x, y) .

Final Exam Review

• Then

$$\begin{aligned} 0 &\leq \frac{x^4 + y^4}{x^2 + y^2} \\ &= \frac{r^4 [\cos^4(\theta) + \sin^4(\theta)]}{r^2} \\ &= r^2 [\cos^4(\theta) + \sin^4(\theta)] \\ &\leq 2r^2 \\ &= 2(x^2 + y^2) \\ &= 2\|(x, y)\|^2. \end{aligned}$$

$$|\cos^4\theta + \sin^4\theta| \leq |\cos^4\theta| + |\sin^4\theta| = 1 + 1 = 2$$

bounded between 0 and 2 $\rightarrow$ Δ inequality

• As $\lim_{(x,y) \rightarrow (0,0)} 2\|(x,y)\|^2 = 0$, the Squeeze Theorem yields $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 + y^4}{x^2 + y^2} = 0$.

• Therefore, f is totally differentiable at $(0,0)$. ✖

Example

Define a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) \stackrel{\text{df}}{=} \begin{cases} \frac{x^3 - xy^2}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

Problem: Determine whether or not f is totally differentiable at $(0, 0)$.

Solution

- For all $x \in \mathbb{R} \setminus \{0\}$, we have

$$f(x, 0) = \frac{x^3 - x \cdot 0^2}{x^2 + 0^2} = \frac{x^3}{x^2} = x.$$

- As $f(0, 0) = 0$, we thus have, for all $x \in \mathbb{R}$, that $f(x, 0) = x$.

- Hence,

$$f_x(0, 0) = \left. \frac{d}{dx}[x] \right|_{x=0} = 1.$$

- For all $y \in \mathbb{R} \setminus \{0\}$, we have

$$f(0, y) = \frac{0^3 + 0y^2}{0^2 + y^2} = \frac{0}{y^2} = 0.$$

- As $f(0, 0) = 0$, we thus have, for all $y \in \mathbb{R}$, that $f(0, y) = 0$.

Final Exam Review

- Hence,

$$f_y(0,0) = \frac{d}{dy}[0] \Big|_{y=0} = 0.$$

- It follows that the tangent-plane function of f at $(0,0)$, denoted by $L_{f,(0,0)}$, exists.
- For all $x, y \in \mathbb{R}$, we have

$$\begin{aligned} L_{f,(0,0)}(x,y) &= f(0,0) + f_x(0,0) \cdot (x-0) + f_y(0,0) \cdot (y-0) \\ &= 0 + 1 \cdot x + 0 \cdot y \\ &= x. \end{aligned}$$

- It now suffices to determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{|f(x,y) - L_{f,(0,0)}(x,y)|}{\|(x,y) - (0,0)\|} = 0$.

This is precisely what it means for f to be totally differentiable at $(0,0)$.

- Let $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

- Then

$$\begin{aligned} \frac{|f(x,y) - L_{f,(0,0)}(x,y)|}{\|(x,y) - (0,0)\|} &= \frac{|f(x,y) - x|}{\|(x,y)\|} \\ &= \frac{\left| \frac{x^3 - xy^2}{x^2 + y^2} - x \right|}{\sqrt{x^2 + y^2}} \\ &= \frac{\left| \frac{x^3 - xy^2}{x^2 + y^2} - \frac{x(x^2 + y^2)}{x^2 + y^2} \right|}{\sqrt{x^2 + y^2}} \\ &= \frac{\left| \frac{x^3 - xy^2}{x^2 + y^2} - \frac{x^3 + xy^2}{x^2 + y^2} \right|}{\sqrt{x^2 + y^2}} \\ &= \frac{\left| -\frac{2xy^2}{x^2 + y^2} \right|}{\sqrt{x^2 + y^2}} \end{aligned}$$

Final Exam Review

$$= \frac{2|x|y^2}{\sqrt{(x^2+y^2)^3}}.$$

- Suspicion: $\lim_{(x,y) \rightarrow (0,0)} \frac{2|x|y^2}{\sqrt{(x^2+y^2)^3}}$ does NOT exist.

- Restrict the limit to the x -axis:

$$\lim_{x \rightarrow 0} \frac{2|x|0^2}{\sqrt{(x^2+0^2)^3}} = \lim_{x \rightarrow 0} \frac{0}{\sqrt{x^6}} = \lim_{x \rightarrow 0} 0 = 0.$$

- Restrict the limit to the line given by the equation $y = x$:

$$\lim_{x \rightarrow 0} \frac{2|x|x^2}{\sqrt{(x^2+x^2)^3}} = \lim_{x \rightarrow 0} \frac{2|x|^3}{\sqrt{(2x^2)^3}} = \lim_{x \rightarrow 0} \frac{2|x|^3}{\sqrt{8x^6}} = \lim_{x \rightarrow 0} \frac{2|x|^3}{2\sqrt{2}|x|^3} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

- As the two restricted limits exist but are different, we find that $\lim_{(x,y) \rightarrow (0,0)} \frac{2|x|y^2}{\sqrt{x^2+y^2}}$ does NOT exist.

- Therefore, f is NOT totally differentiable at $(0,0)$. ✖